

# FPROS1-A1620-OSB

Serie FPROS • R1234ZE • senza serbatoio / without tank



FPROS1-A1620-OSB

RESA FRIGORIFERA • COOLING CAPACITY

1.053 kW

EER • 12/7 °C • AIR 35 °C

2,98

RESA NOMINALE • 15 °C • AIR 25 °C

1.614 kW

PORTATA ACQUA • WATER FLOW

181,1 m<sup>3</sup>/h

|                            |                 |                                        |              |
|----------------------------|-----------------|----------------------------------------|--------------|
| Compressori / Compressors  | 2 × vite        | L × P × H (mm)                         | su richiesta |
| Circuiti / Circuits        | 2               | Serbatoio / Tank                       | —            |
| Ventilatori / Fans         | 26              | Taglia / Size                          | 1620         |
| Free cooling (ΔT 8 K)      | 1.200 kW *      | Campo aria esterna / Ambient air range | -5 / +47 °C  |
| Refrigerante / Refrigerant | R1234ZE • GWP 7 |                                        |              |

## Resa in sintesi / Capacity at a glance kW

| ARIA ESTERNA / AMBIENT AIR                  | 25 °C | 30 °C | 35 °C | 40 °C | 45 °C |
|---------------------------------------------|-------|-------|-------|-------|-------|
| Acqua in uscita 7 °C / Leaving water 7 °C   | 1.183 | 1.118 | 1.053 | 987   | 919   |
| Acqua in uscita 15 °C / Leaving water 15 °C | 1.614 | 1.528 | 1.443 | 1.356 | 1.269 |

Refrigeratori con free cooling integrato e compressori a vite, per grandi potenze che lavorano tutto l'anno.

Chillers with integrated free cooling and screw compressors, for large capacities running all year round.

# Dati tecnici

## Technical data

FPROS1 - A1620 - OSB

**Prestazioni nominali — acqua in uscita 15 °C, aria 25 °C / Nominal performance — 15 °C leaving water, 25 °C air**

|                                     |                          |
|-------------------------------------|--------------------------|
| Resa frigorifera / Cooling capacity | 1.613,56 kW              |
| Resa frigorifera / Cooling capacity | 1.387.662 kcal/h         |
| Potenza assorbita / Power input     | 303,58 kW                |
| EER / EER                           | 5,32                     |
| Portata acqua / Water flow          | 277,53 m <sup>3</sup> /h |

**Condizioni standard — acqua 12/7 °C, aria 35 °C / Standard conditions — 12/7 °C water, 35 °C air**

|                                     |                          |
|-------------------------------------|--------------------------|
| Resa frigorifera / Cooling capacity | 1.052,79 kW              |
| Potenza assorbita / Power input     | 353,76 kW                |
| EER / EER                           | 2,98                     |
| ESEER / ESEER                       | 3,96                     |
| Portata acqua / Water flow          | 181,08 m <sup>3</sup> /h |

### Componenti / Components

|                                               |                       |
|-----------------------------------------------|-----------------------|
| Compressori / Compressors                     | 2 × Vite              |
| Circuiti frigoriferi / Refrigerant circuits   | 2                     |
| Ventilatori / Fans                            | 26                    |
| Free cooling (ΔT 8 K) / Free cooling (8 K ΔT) | 1.200 kW *            |
| Ventilatori free cooler / Free cooler fans    | 28                    |
| Condensatore / Condenser                      | a microcanali         |
| Evaporatore / Evaporator                      | a fascio tubiero      |
| Refrigerante / Refrigerant                    | R1234ZE • GWP 7 • A2L |

### Installazione / Installation

|                                                                                           |                           |
|-------------------------------------------------------------------------------------------|---------------------------|
| Dimensioni L × P × H / Dimensions L × D × H                                               | su richiesta / on request |
| Campo aria esterna / Ambient air range                                                    | -5 / +47 °C               |
| Dati elettrici, carica gas, peso, attacchi / Electrical data, charge, weight, connections | su richiesta / on request |

\* Potenza di free cooling indicativa, da confermare in fase di offerta. / Indicative free cooling capacity, to be confirmed at quotation stage.

### Dotazioni di serie / Standard equipment

- Valvole di intercettazione  
Shut-off valves
- Griglia di protezione condensatore  
Condenser protection grille
- Valvola di espansione elettronica  
Electronic expansion valve
- Condensatore a microcanali  
Micro-channel condenser
- Evaporatore a fascio tubiero  
Shell & tube evaporator
- Display 4"/7"  
4"/7" display
- Pressostati e manometri  
Pressure switches and gauges
- Pressostato differenziale acqua  
Water differential pressure switch
- Presa di servizio 220 V e luce quadro elettrico  
220 V service socket and electrical panel light

### Con glicole / With glycol

| MISCELA  | COEFF.  | 12/7 • 35 °C |
|----------|---------|--------------|
| MEG 20 % | × 0,980 | 1.031,7 kW   |
| MEG 30 % | × 0,950 | 1.000,2 kW   |
| MEG 40 % | × 0,920 | 968,6 kW     |
| MPG 20 % | × 0,975 | 1.026,5 kW   |
| MPG 30 % | × 0,960 | 1.010,7 kW   |
| MPG 40 % | × 0,950 | 1.000,2 kW   |

# Prestazioni Performance

FPROS1-A1620-OSB · acqua pura,  $\Delta T$  5 K · pure water, 5 K  $\Delta T$  ·  $\square$  12/7 °C · 35 °C

## Resa frigorifera / Cooling capacity (kW)

| LWT ↓ · air → | 15    | 20    | 25    | 30    | 35    | 40    | 45    |
|---------------|-------|-------|-------|-------|-------|-------|-------|
| 1             | 1.024 | 974   | 923   | 870   | 818   | 764   | 710   |
| 2             | 1.069 | 1.016 | 964   | 909   | 854   | 799   | 743   |
| 3             | 1.113 | 1.059 | 1.004 | 948   | 891   | 833   | 775   |
| 4             | 1.160 | 1.104 | 1.048 | 989   | 930   | 870   | 810   |
| 5             | 1.207 | 1.149 | 1.091 | 1.030 | 969   | 907   | 845   |
| 6             | 1.258 | 1.197 | 1.137 | 1.074 | 1.011 | 947   | 882   |
| 7             | 1.308 | 1.246 | 1.183 | 1.118 | 1.053 | 987   | 919   |
| 8             | 1.362 | 1.297 | 1.232 | 1.165 | 1.097 | 1.029 | 959   |
| 9             | 1.415 | 1.348 | 1.281 | 1.211 | 1.142 | 1.071 | 999   |
| 10            | 1.472 | 1.403 | 1.333 | 1.261 | 1.189 | 1.115 | 1.041 |
| 11            | 1.529 | 1.457 | 1.386 | 1.311 | 1.236 | 1.160 | 1.083 |
| 12            | 1.589 | 1.515 | 1.441 | 1.364 | 1.286 | 1.208 | 1.128 |
| 13            | 1.649 | 1.573 | 1.496 | 1.416 | 1.337 | 1.255 | 1.173 |
| 14            | 1.713 | 1.634 | 1.555 | 1.472 | 1.390 | 1.306 | 1.221 |
| 15            | 1.777 | 1.695 | 1.614 | 1.528 | 1.443 | 1.356 | 1.269 |

## Potenza assorbita / Power input (kW)

| LWT ↓ · air → | 15  | 20  | 25  | 30  | 35  | 40  | 45  |
|---------------|-----|-----|-----|-----|-----|-----|-----|
| 1             | 245 | 267 | 291 | 318 | 348 | 380 | 416 |
| 2             | 246 | 268 | 292 | 319 | 349 | 382 | 418 |
| 3             | 247 | 269 | 293 | 320 | 350 | 383 | 419 |
| 4             | 248 | 270 | 294 | 321 | 351 | 384 | 420 |
| 5             | 248 | 270 | 295 | 322 | 352 | 385 | 421 |
| 6             | 249 | 271 | 296 | 323 | 353 | 386 | 423 |
| 7             | 250 | 272 | 296 | 324 | 354 | 387 | 424 |
| 8             | 251 | 273 | 297 | 324 | 355 | 388 | 425 |
| 9             | 252 | 274 | 298 | 325 | 356 | 389 | 426 |
| 10            | 253 | 275 | 299 | 326 | 356 | 390 | 427 |
| 11            | 254 | 276 | 300 | 327 | 357 | 391 | 428 |
| 12            | 255 | 277 | 301 | 328 | 358 | 392 | 429 |
| 13            | 256 | 277 | 302 | 329 | 359 | 393 | 430 |
| 14            | 257 | 278 | 303 | 330 | 360 | 394 | 431 |
| 15            | 258 | 279 | 304 | 330 | 361 | 395 | 432 |

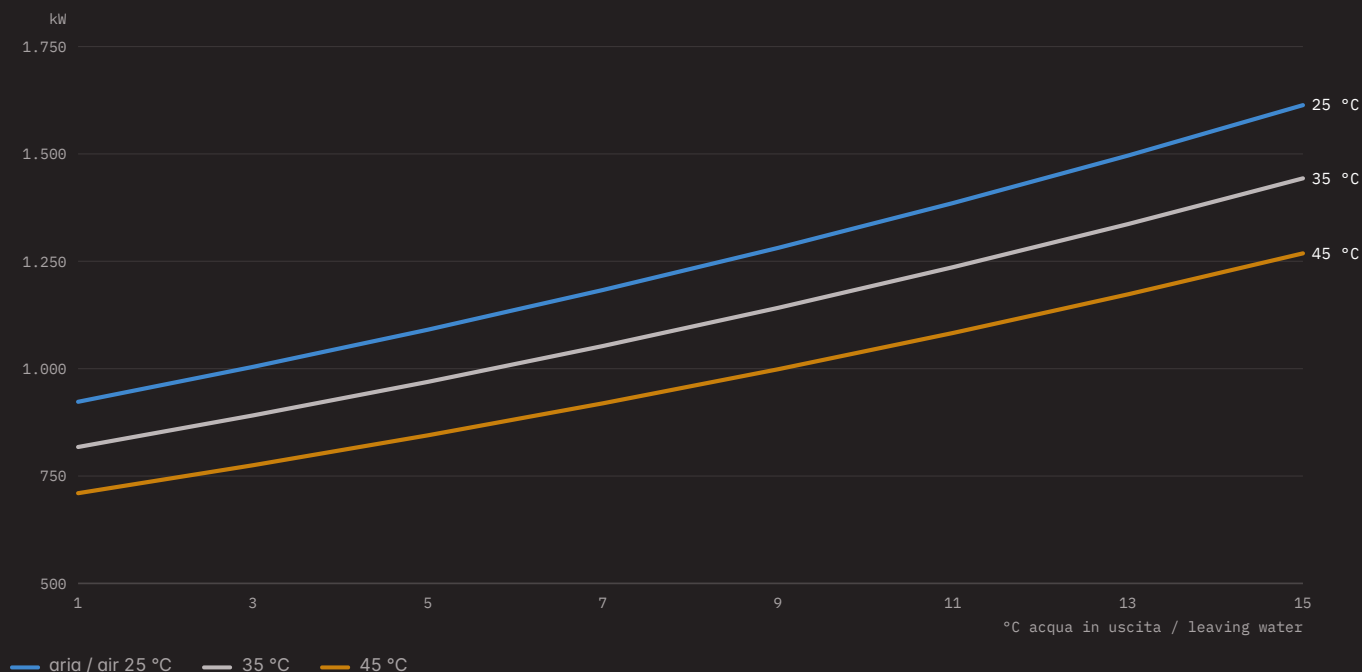
## EER / EER (kW/kW)

| LWT ↓ · air → | 15   | 20   | 25   | 30   | 35   | 40   | 45   |
|---------------|------|------|------|------|------|------|------|
| 1             | 4,18 | 3,65 | 3,17 | 2,74 | 2,35 | 2,01 | 1,71 |
| 2             | 4,35 | 3,79 | 3,30 | 2,85 | 2,45 | 2,09 | 1,78 |
| 3             | 4,51 | 3,94 | 3,43 | 2,96 | 2,55 | 2,18 | 1,85 |
| 4             | 4,69 | 4,09 | 3,56 | 3,08 | 2,65 | 2,27 | 1,93 |
| 5             | 4,86 | 4,25 | 3,70 | 3,20 | 2,76 | 2,36 | 2,00 |
| 6             | 5,05 | 4,41 | 3,85 | 3,33 | 2,87 | 2,45 | 2,09 |
| 7             | 5,23 | 4,58 | 3,99 | 3,46 | 2,98 | 2,55 | 2,17 |
| 8             | 5,42 | 4,75 | 4,14 | 3,59 | 3,09 | 2,65 | 2,26 |
| 9             | 5,62 | 4,92 | 4,30 | 3,73 | 3,21 | 2,75 | 2,34 |
| 10            | 5,82 | 5,11 | 4,46 | 3,87 | 3,34 | 2,86 | 2,44 |
| 11            | 6,02 | 5,29 | 4,62 | 4,01 | 3,46 | 2,97 | 2,53 |
| 12            | 6,24 | 5,48 | 4,79 | 4,16 | 3,59 | 3,08 | 2,63 |
| 13            | 6,45 | 5,67 | 4,96 | 4,31 | 3,72 | 3,20 | 2,73 |
| 14            | 6,67 | 5,87 | 5,14 | 4,47 | 3,86 | 3,32 | 2,83 |
| 15            | 6,89 | 6,07 | 5,32 | 4,63 | 4,00 | 3,44 | 2,94 |

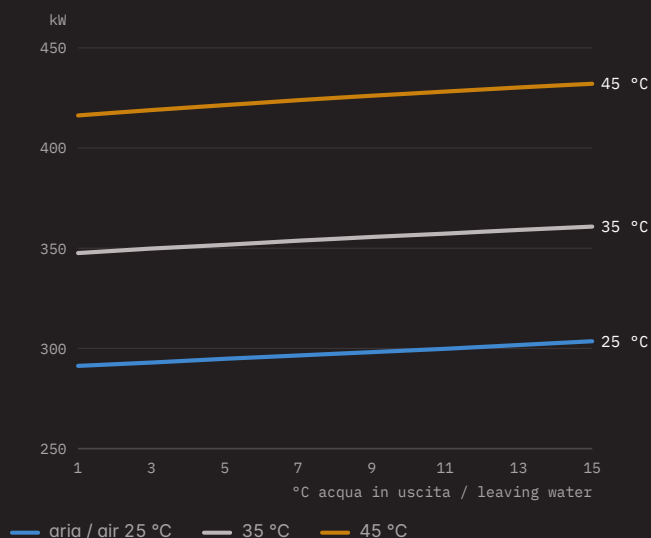
# Curve Curves

FPROS1-A1620-OSB · acqua pura,  $\Delta T$  5 K · pure water, 5 K  $\Delta T$

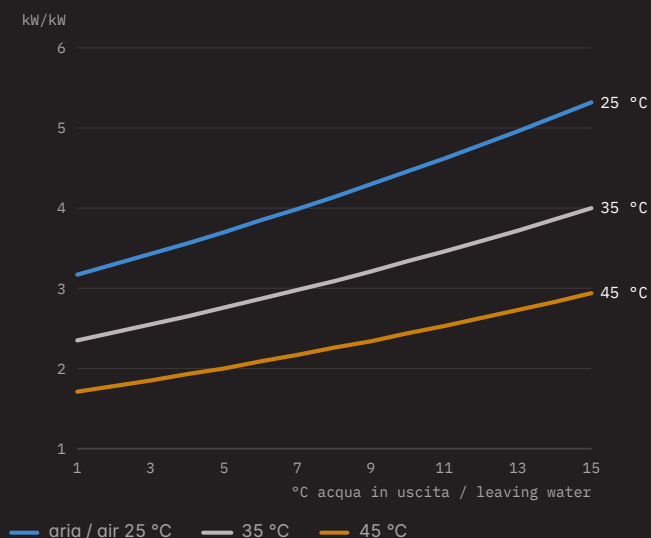
## Resa frigorifera / Cooling capacity



## Potenza assorbita / Power input



## EER / EER



Dimensioni, pesi e disegno d'ingombro sono forniti su richiesta per la configurazione scelta. / Dimensions, weights and dimensional drawing are available on request for the chosen configuration.

# Optional — 1/2 Options

FPROS1-A1620-OSB · 55 optional disponibili per questa configurazione · 55 options available for this configuration

## Pompe / Pumps

Pompa 214 m<sup>3</sup>/h – prevalenza 25 m

Pump 214 m<sup>3</sup>/h – head 25 m

Pompa 214 m<sup>3</sup>/h – prevalenza 40 m

Pump 214 m<sup>3</sup>/h – head 40 m

Pompa doppia montata 214 m<sup>3</sup>/h – prevalenza 25 m

Twin pump, factory -fitted, 214 m<sup>3</sup>/h – head 25 m

Pompa doppia montata 214 m<sup>3</sup>/h – prevalenza 40 m

Twin pump, factory -fitted, 214 m<sup>3</sup>/h – head 40 m

## Serbatoi e vasi di espansione / Tanks and expansion vessels

Serbatoio idraulico esterno 3000 l in AISI 304 con pompa doppia in AISI 304 montata

External water tank 3000 l, AISI 304 with factory-fitted twin AISI 304 pump

Serbatoio idraulico esterno 5000 l in AISI 304 con pompa doppia in AISI 304 montata

External water tank 5000 l, AISI 304 with factory-fitted twin AISI 304 pump

Serbatoio idraulico esterno 3000 l in AISI 304 con pompa doppia montata

External water tank 3000 l, AISI 304 with factory-fitted twin pump

Serbatoio idraulico esterno 5000 l in AISI 304 con pompa doppia montata

External water tank 5000 l, AISI 304 with factory-fitted twin pump

Serbatoio idraulico esterno 3000 l in AISI 304 con pompa singola in AISI 304 montata

External water tank 3000 l, AISI 304 with factory-fitted single AISI 304 pump

Serbatoio idraulico esterno 5000 l in AISI 304 con pompa singola in AISI 304 montata

External water tank 5000 l, AISI 304 with factory-fitted single AISI 304 pump

Serbatoio idraulico esterno 3000 l in AISI 304 con pompa singola montata

External water tank 3000 l, AISI 304 with factory-fitted single pump

Serbatoio idraulico esterno 5000 l in AISI 304 con pompa singola montata

External water tank 5000 l, AISI 304 with factory-fitted single pump

Serbatoio idraulico esterno 3000 l pressurizzato in acciaio zincato con pompa singola montata

External water tank 3000 l, pressurised, galvanised steel with factory-fitted single pump

Serbatoio idraulico esterno 5000 l pressurizzato in acciaio zincato con pompa singola montata

External water tank 5000 l, pressurised, galvanised steel with factory-fitted single pump

Vaso di espansione 100 l

Expansion vessel 100 l

Vaso di espansione 200 l

Expansion vessel 200 l

Vaso di espansione 300 l

Expansion vessel 300 l

Vaso di espansione 500 l

Expansion vessel 500 l

Vaso di espansione 750 l

Expansion vessel 750 l

## Valvole e filtri / Valves and filters

Elettrovalvola 2"

Solenoid valve 2"

Elettrovalvola 3"

Solenoid valve 3"

Elettrovalvola 4"

Solenoid valve 4"

Elettrovalvola 5"

Solenoid valve 5"

Elettrovalvola 6"

Solenoid valve 6"

Filtro a dischi in acciaio 2"

Steel disc filter 2"

Filtro a dischi in acciaio 3"

Steel disc filter 3"

Filtro a dischi in acciaio 4"

Steel disc filter 4"

Filtro meccanico a Y 2"

Y-strainer 2"

Filtro meccanico a Y 3"

Y-strainer 3"

Filtro meccanico a Y 4"

Y-strainer 4"

Filtro meccanico a Y 5"

Y-strainer 5"

Filtro meccanico a Y 6"

Y-strainer 6"

Valvola a 3 vie 2" miscelatrice

3-way mixing valve 2"

Valvola a 3 vie 3" miscelatrice

3-way mixing valve 3"

Valvola a 3 vie 4" miscelatrice

3-way mixing valve 4"

Valvola a 3 vie 5" miscelatrice

3-way mixing valve 5"

Valvola a 3 vie 6" miscelatrice

3-way mixing valve 6"

Valvola a 3 vie 2" on/off

3-way valve 2" on/off

Valvola a 3 vie 3" on/off

3-way valve 3" on/off

Valvola a 3 vie 4" on/off

3-way valve 4" on/off

Valvola a 3 vie 5" on/off

3-way valve 5" on/off

# Optional — 2/2

## Options

FPROS1-A1620-OSB · 55 optional disponibili per questa configurazione · 55 options available for this configuration

### Valvole e filtri / Valves and filters

Valvola a 3 vie 6" on/off  
3-way valve 6" on/off

### Circuito frigorifero / Refrigerant circuit

Cofanatura insonorizzante compressori  
Compressor acoustic enclosure

Compressore inverter  
Inverter compressor

Desurriscaldatore a piastre  
Plate desuperheater

Rilevatore fughe gas  
Refrigerant leak detector

### Ventilazione e clima / Fans and climate

Inverter ventilatori  
Fan inverter

Ventilatori EC  
EC fans

### Controllo e comunicazione / Control and communication

Comando remoto con display (cavo escluso)  
Remote control with display (cable not included)

Comunicazione RS485 Modbus RTU / TCP-IP (cavo escluso)  
RS485 Modbus RTU / TCP-IP communication (cable not included)

Comunicazione RS485 verso BACnet (cavo escluso)  
RS485 to BACnet communication (cable not included)

### Quadro elettrico / Electrical panel

Analizzatore di rete  
Power analyser

Kit invernale (inverter ventilatori + resistenza quadro + resistenza evaporatore + termostato) fino a -25 °C  
Winter kit (fan inverter + panel heater + evaporator heater + thermostat) down to -25 °C

### Installazione / Installation

Antivibranti a molla  
Spring anti-vibration mounts

Antivibranti in gomma  
Rubber anti-vibration mounts

## Note Notes

- Prestazioni con acqua pura, salto termico 5 K. Le condizioni di riferimento sono indicate in ogni tabella.
  - Con miscele di glicole la resa frigorifera si corregge con i coefficienti della tabella dedicata; potenza assorbita ed EER restano invariati.
  - Dimensioni ricavate dai disegni quotati (tolleranza  $\pm 10$  mm): da confermare in fase d'ordine.
  - Dati elettrici, carica di refrigerante, pesi e attacchi idraulici sono forniti su richiesta per la configurazione scelta.
  - Procryon si riserva di modificare dati e caratteristiche senza preavviso, per il miglioramento continuo del prodotto.
  - Free cooling: potenza riferita a una differenza di 8 K tra acqua e aria esterna; valore indicativo, da confermare in fase di offerta.
- Performance with pure water, 5 K temperature difference. Reference conditions are stated in each table.
  - With glycol mixtures, cooling capacity is corrected with the coefficients in the dedicated table; power input and EER are unchanged.
  - Dimensions taken from dimensional drawings (tolerance  $\pm 10$  mm): to be confirmed at order stage.
  - Electrical data, refrigerant charge, weights and water connections are available on request for the chosen configuration.
  - Procryon reserves the right to change data and features without notice, as part of continuous product improvement.
  - Free cooling: capacity referred to an 8 K difference between water and outdoor air; indicative value, to be confirmed at quotation stage.

## Parliamoci Let's talk

Progettazione, intervento urgente, valutazione di un impianto esistente, supporto tecnico e intervento rapido — 24/7.

System design, urgent response, assessment of an existing plant, technical support and rapid response — 24/7.

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Continuity by design